

Gout & Uric Acid in Kidney Disease

Acute flares, diet management, uric acid lowering medications, Filipino food context, and kidney protection. Written for patients with gout, CKD, and **hyperuricemia**.



W.G.M. Rivero MD
FPCP · DPSN
Nephrologist

williamriveromd.com

<6.0 mg/dL · Uric Acid Target <i>Uric acid > 6.8 mg/dL → crystals form in joints, kidneys, and soft tissues. Keeping uric acid < 6.0 mg/dL on treatment prevents flares and protects the kidneys.</i>	↑Risk CKD Progression	Allopurinol Primary Treatment	Diet Modifies Risk
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1 What Is Uric Acid — and Why Does It Build Up?

Uric acid is the normal **end product of purine metabolism**. Purines are broken down in the liver by an enzyme called xanthine oxidase. About 70% of uric acid is excreted by the kidneys in urine; the remainder goes through the gut. When production exceeds excretion — due to diet, genetics, or reduced kidney function — serum uric acid rises (hyperuricemia). Above **6.8 mg/dL**, uric acid is supersaturated in blood and begins forming needle-shaped monosodium urate (MSU) crystals in cooler peripheral joints, tendon sheaths, and renal tubules.

Uric Acid Targets (on treatment)

Men: < 6.0 mg/dL (general); < 5.0 mg/dL if tophi present

Women: < 5.0 mg/dL (post-menopausal = same as men)

CKD patients: < 6.0 mg/dL — KDIGO 2024

Saturation point: 6.8 mg/dL — crystals form above this

Do NOT target zero — uric acid is a natural antioxidant

The Gout–CKD Vicious Cycle

High uric acid → deposits crystals in renal tubules (urate nephropathy)

↓

Kidney function declines → less uric acid excreted in urine

↓

Serum uric acid rises further → more crystal deposition → more CKD

↓

Breaking this cycle requires uric acid lowering therapy early

2 Gout Spectrum — Acute Flare vs Chronic Tophaceous Gout

Acute Gout Flare

Sudden onset of severe joint pain — usually at night

Typical joint: First metatarsophalangeal joint (big toe) — "podagra." Also: ankle, knee, wrist, elbow.

Onset: Maximal pain within 6–12 hours. Skin red, hot, swollen, exquisitely tender — patient cannot bear bedsheet contact.

Duration: Untreated: 7–14 days. Treated early: 3–5 days.

Triggers: Dehydration, alcohol (especially beer), high-purine meal, rapid uric acid change (starting allopurinol), infection, surgery, diuretics.

Chronic Tophaceous Gout

Years of untreated hyperuricemia → crystal deposits in soft tissue

Tophi: Chalky white deposits visible under skin — ear helix, knuckles, Achilles tendon, elbow bursae. Can ulcerate and become infected.

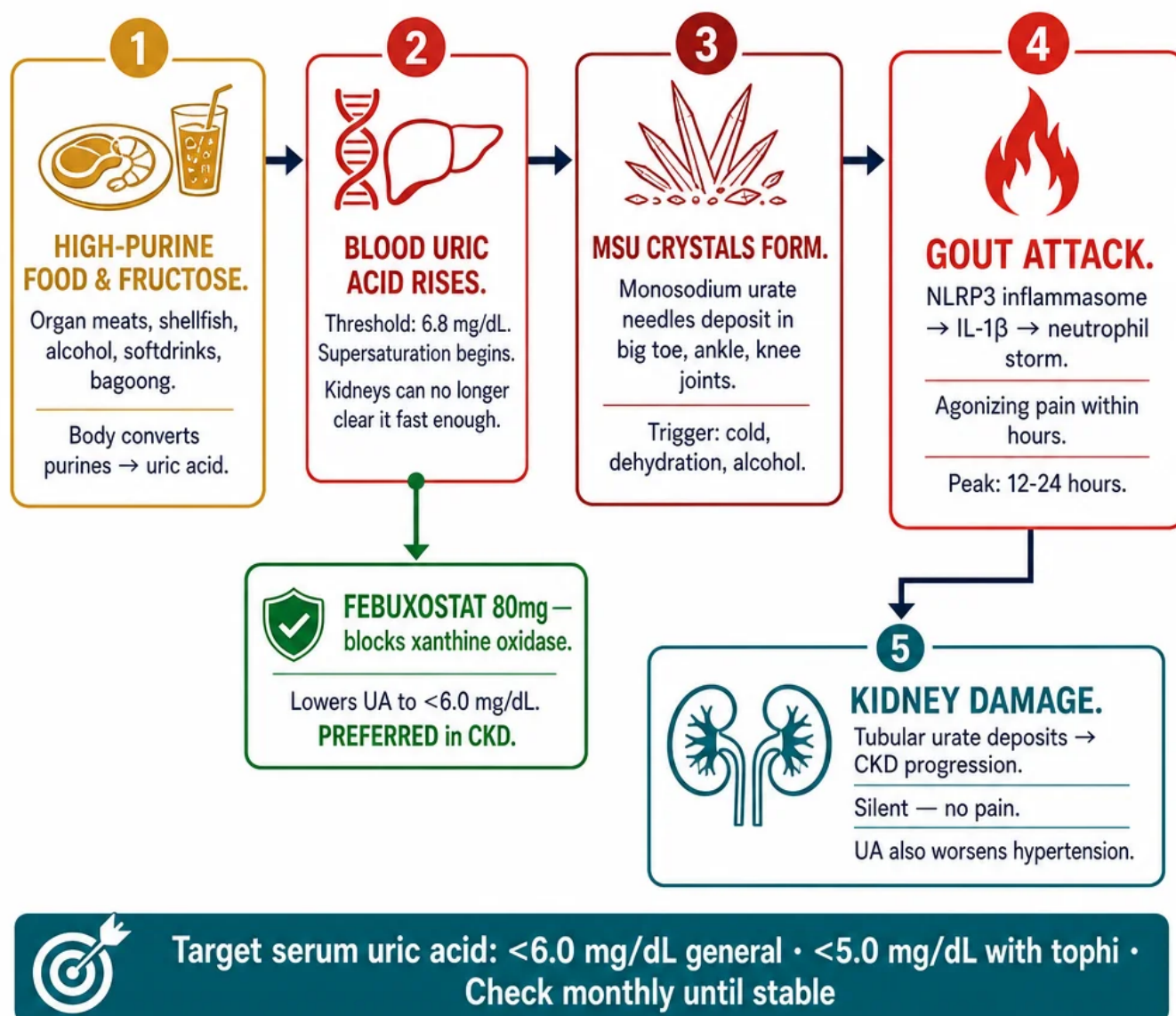
Joint damage: Chronic erosive arthropathy — permanent joint destruction resembling RA on X-ray. Punched-out erosions with overhanging edges.

Kidney impact: Urate nephropathy (tubular crystal deposits), uric acid kidney stones, CKD progression, hypertension.

Treatment: Requires long-term uric acid lowering therapy (ULT) — allopurinol or febuxostat — for life.

WHAT IS URIC ACID — PURINE METABOLISM & CKD

HOW URIC ACID BUILDS UP AND ATTACKS YOUR JOINTS



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Fig. 1 — Uric acid is the end product of purine metabolism. In CKD, impaired kidney excretion causes serum uric acid to accumulate, leading to crystal deposition in joints, soft tissues, and tubules. The vicious cycle: high uric acid damages kidneys → impaired excretion → higher uric acid. Every CKD patient should have serum uric acid measured and treated if > 6.0 mg/dL.

THE INFLAMMATORY CASCADE — HOW A GOUT ATTACK HAPPENS

How Uric Acid Crystals Cause a Gout Attack

A gout flare is not simply “high uric acid” — it is an acute inflammatory response triggered when monosodium urate (MSU) crystals are recognized by immune cells in the joint fluid.

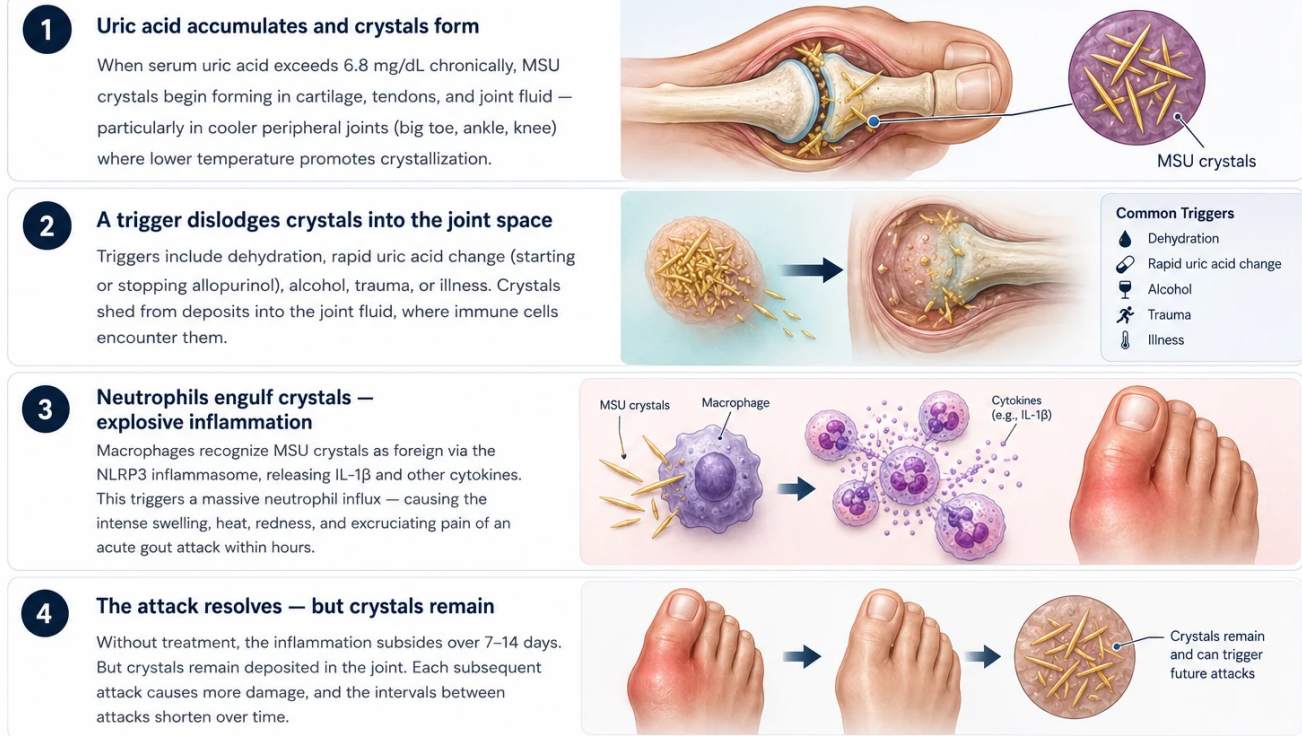


Fig. 2 — The inflammatory cascade of an acute gout attack: monosodium urate crystals deposit in synovial fluid → macrophages recognize crystals via the NLRP3 inflammasome → IL-1 β released → massive neutrophil influx → intense joint inflammation, warmth, swelling, and severe pain within 6–12 hours. Each untreated attack causes cumulative joint erosion. Treating early (within 12–24 hours) dramatically shortens the flare.

Food (Filipino name)	Purines	Guidance
VERY HIGH (>300 MG/100G) — AVOID COMPLETELY DURING FLARE; SEVERELY LIMIT OTHERWISE		
Dilis (dried anchovies)	~600 mg	Extreme — do not eat during a flare; even small amounts can trigger attacks
Sardinas (canned, with bones)	~480 mg	Extreme — avoid; hidden in sinigang base packets and pasta sauces
Atay ng baka / manok (liver)	470–550 mg	Extreme — the single worst food trigger for gout in Filipino diet
Bato ng baka (kidney), Dinuguan	480–510 mg	Extreme — organ meats raise uric acid more than any other food group
Sweetbreads (thymus/pancreas)	~825 mg	Extreme — highest purine food known; never eat with gout
Goto / tripe (callos)	~470 mg	Extreme — popular breakfast; a bowl can trigger a flare within 24 hours
Tuyo (dried fish)	~290 mg	Very high — dehydrated fish concentrates purines; even 30 g is risky
Beer (any brand, 330 mL)	Doubles flare risk	Beer contains guanosine (a purine) + ethanol both inhibit uric acid excretion — worst single trigger
Softdrinks with HFCS (Coke, Pepsi, Tang, Zest-O)	Raises UA 0.3–0.5 mg/dL	Fructose → directly metabolized to uric acid in liver; one 500 mL bottle can trigger a flare
Bagoong alamang / isda (1 tbsp)	~250 mg	Hidden trigger in kare-kare, pinakbet, dipping — concentrated shrimp purines
MODERATE–HIGH (50–300 MG/100G) — LIMIT TO 1 SERVING/DAY; AVOID DURING FLARE		
Hipon / sugpo (shrimp/prawns, 100 g)	~265 mg	High — 1 small serving occasionally; avoid during acute flare
Pusit (squid, 100 g)	~190 mg	High — limit to 1 serving/week maximum
Alimango / alimasag (crab, 100 g)	~150 mg	High — occasional small serving; crabfat (aligue) especially rich in purines
Bangus (milkfish, grilled, 100 g)	~165 mg	High — limit to ½ serving; tilapia is a safer fish choice
Manok / baboy / baka (100 g)	120–175 mg	Moderate-high — 1 palm-sized serving/day OK; avoid organ parts
Tilapia (grilled, 100 g)	~120 mg	Moderate — the safest fish choice for gout patients; 1 serving/day OK
Monggo (½ cup cooked)	~68 mg	Moderate — but plant purines carry LOWER flare risk than animal purines; can eat 3–4× week
Mushrooms / kabute (1 cup cooked)	~60 mg	Moderate — plant purine; no increased flare risk in studies; safe to eat regularly
Kangkong, cauliflower (1 cup)	40–50 mg	Moderate plant purine — SAFE to eat freely; plant purines do not raise flare risk
LOW / SAFE — EAT FREELY (UNLESS RESTRICTED FOR OTHER MEDICAL REASONS)		
Itlog (eggs) — 1 large	~5 mg	Best animal protein for gout patients — very low purine, highly nutritious
Tokwa / tofu (100 g)	~70 mg	Safe — plant purines; studies show tofu may actually lower uric acid over time
Kanin / rice, tinapay, kamote, cassava	<15 mg	Safe — staple carbohydrates do not raise uric acid
Most fruits (papaya, apple, pear, saging)	<10 mg	Safe — eat whole; avoid sweetened/canned; cherries may actually lower uric acid
Kape (coffee, 1–2 cups brewed)	Protective	Studies: 1–2 cups/day lowers serum uric acid by 0.1–0.4 mg/dL; mechanism unclear
Low-fat milk, plain yogurt (1 cup)	Protective	Orotic acid in dairy promotes uric acid excretion — lowers UA; safe daily
Tubig / plain water (2–3 L/day)	Protective	Most important non-drug intervention — hydration improves uric acid excretion

KEY: Plant purines (vegetables, legumes, tofu) have significantly lower gout flare risk than animal purines (organ meats, shellfish, red meat) despite similar purine content. Fructose (from HFCS in softdrinks) is metabolized directly to uric acid — a major underrecognized trigger.

GOUT-FRIENDLY FILIPINO DIET — WHAT TO EAT AND WHAT TO AVOID

What to Eat — and What Triggers Flares

Diet alone cannot normalize uric acid in most patients with established gout, but it significantly reduces flare frequency and supports medication effectiveness. In Filipino patients, the biggest dietary drivers are organ meats, shellfish, beer, and softdrinks with fructose.



Always avoid — high flare risk



Organ meats: liver, kidney, brain, intestines (goto, dinuguan)



Beer and spirits (especially dark beer and gin)



Sardines, anchovies, mackerel (high-purine fish)



Softdrinks with high-fructose corn syrup (regular Coke, Sprite)



Sugary fruit juices (fructose raises uric acid independently of purines)



Bagoong, shrimp paste, fermented fish products



Safe — eat freely



Eggs (very low purine — excellent protein source)



Low-fat dairy: milk, yogurt (actually lowers uric acid)



Most vegetables including camote, kangkong, pechay



White rice, bread, pasta, corn



Coffee (associated with lower uric acid levels)



Water — aim for 2–3 liters daily to promote uric acid excretion



Cherries and tart cherry juice (modest anti-inflammatory effect)



Moderate — limit portions



Red meat (beef, pork): maximum 1 serving/day



Shellfish (tahong, halaaan, hipon): once or twice weekly



Bangus and tilapia: moderate portions acceptable



Tofu and legumes (mongo, tokwa): small-to-moderate portions despite purines — overall effect is neutral or beneficial



Wine: one glass with a meal is much less harmful than beer



Protective foods — include regularly



Low-fat milk and yogurt (increases uric acid excretion)



Vitamin C-rich foods: calamansi, guava, papaya (mild uricosuric effect)



Ampalaya (bitter melon) — modest xanthine oxidase inhibition



Adequate hydration — the single most impactful dietary measure



Alkaline foods (vegetables, fruits) — raise urine pH, reduce stone risk



Key takeaway: Focus on whole foods, healthy portions, and hydration. Combine with proper medications for optimal gout control.



Hydrate



Watch portions



Eat balanced



Take meds

Fig. 3 — Gout-friendly Filipino diet: emphasize rice, eggs, low-fat protein, fruits, and water. Avoid organ meats (atay, goto, dinuguan), dried fish (dilis, tuyo, sardinas), beer, and high-fructose drinks (Coke, Pepsi, sweetened juices). Tofu and low-fat dairy may actually lower uric acid levels. Diet alone cannot normalize uric acid in established gout, but it significantly reduces flare frequency and complements medication.

HIDDEN URIC ACID TRIGGERS IN THE FILIPINO DIET

HIDDEN URIC ACID DRIVERS IN **FILIPINO** FOOD & DAILY LIFE

1 

**FRUCTOSE —
The Silent Driver.**

Softdrinks, juice drinks,
sweetened coffee.
Fructose raises uric acid
as fast as alcohol.
**Switch to water or
unsweetened drinks.**

2 

BAGOONG & PATIS.

Fermented shrimp paste
and fish sauce are
extremely high-purine.
**Use calamansi +
herbs instead.**

3 

BEER & GIN.

Alcohol blocks uric acid
excretion AND beer is
high-purine.
**Even one beer can
trigger a flare.**

4 

MONGGO & BEANS.

High in purines but not
as high as meat.
**Limit to 2–3× per week.
Tofu is safer.**

5 

DEHYDRATION.

Less urine → less uric
acid excretion.
**Drink 2–3 liters water
daily. Especially during
hot weather.**

6 

RAPID WEIGHT LOSS.

Crash diets increase
uric acid release from
tissue breakdown.
Lose weight slowly.

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Fig. 4 — Hidden uric acid triggers in the Filipino diet: fruit juice (fructose raises uric acid even from "healthy" drinks like commercial calamansi juice with sugar), processed meat seasonings (Knorr cubes, Maggi liquid seasoning contain hydrolyzed meat/yeast extracts), instant soups and sinigang base packets with fish/shrimp concentrate, and softdrinks with high-fructose corn syrup. Bagoong alamang in kare-kare and pinakbet is another commonly missed trigger.

What to Do During an Acute Gout Flare — 6-Step Protocol

1

Rest the joint

Do NOT walk on a swollen foot. Elevate the affected limb above heart level. Avoid all pressure on the joint for 24–48 hours.

2

Ice, not heat

Apply ice pack 20 min ON, 20 min OFF — repeat 4–6×/day. Never apply heat — it worsens inflammation. Wrap ice in towel to prevent frostbite.

3

Medicate early

Take colchicine or NSAIDs as prescribed — START WITHIN 12–24 HOURS of first pain. Early treatment = shorter, less severe flare.

4

Hydrate

Increase water to 2–3 L/day (if not fluid-restricted by your nephrologist). Good hydration improves uric acid excretion.

5

Stop alcohol

Stop all alcohol immediately — beer especially. Even 1 bottle prolongs the flare significantly. No alcohol until uric acid is controlled.

6

Keep allopurinol

Do NOT stop allopurinol if already prescribed — sudden withdrawal makes flares worse. Continue dose as prescribed through the flare.

CKD Warning — NSAIDs (Mefenamic Acid, Ibuprofen, Diclofenac) Are Dangerous in Kidney Disease

Do NOT take NSAIDs if eGFR < 30 mL/min/1.73m². NSAIDs cause acute kidney injury (AKI) by reducing renal blood flow — a single dose can cause a significant and sometimes permanent drop in kidney function. Ask your nephrologist before taking any pain medication during a flare. Use colchicine with dose reduction instead (see medication table on next page). If colchicine is contraindicated, prednisolone 20–40 mg/day for 3–5 days is the alternative.

When to Go to the Emergency Room (ER) Immediately

Warning sign	Why it is an emergency
Fever > 38.5°C with joint swelling	This may be septic arthritis (joint infection) — not gout — and requires emergency joint aspiration and IV antibiotics. Septic arthritis can destroy a joint in 24 hours.
Tophi breaking through skin	Open tophi are an entry point for infection. Requires wound care, possible IV antibiotics, and urgent uric acid control.
Multiple joints swollen simultaneously	Polyarticular gout flare — may require IV colchicine or systemic steroids. Also rule out other arthritis (RA, reactive arthritis).
No improvement after 5 days of treatment	Consider alternative diagnosis: septic arthritis, pseudogout (calcium pyrophosphate), or inadequate medication dosing.

🕒 How Long Does a Flare Last?

- Untreated: 7–14 days before spontaneous resolution
- Treated within 12 hours: 3–5 days
- Treated within 24 hours: 5–7 days
- Key principle: The earlier the treatment, the shorter and milder the flare. Do not wait hoping it will pass — treat immediately.

Correct Colchicine Dosing in CKD

- eGFR > 50: Standard dose — 1.2 mg at first sign, then 0.6 mg 1 hour later
- eGFR 30–50: Reduce dose — 0.6 mg once at onset; do not repeat within 24h
- eGFR < 30: Use with extreme caution — 0.3 mg once; avoid repeated dosing
- On dialysis: Colchicine generally avoided — use prednisolone instead
- Colchicine interacts with clarithromycin, cyclosporine, and some statins — alert your doctor

ACUTE GOUT FLARE — CLASSIC PRESENTATION & COMMON SITES

ACUTE FLARE MANAGEMENT

What to Do During a Gout Attack

1 Act within 12–24 hours for best results

Anti-inflammatory medications work best when started early. If you have a prior diagnosis of gout and your doctor has given you a rescue medication (colchicine or prednisone), take it at the first sign of a flare — do not wait until the pain is unbearable.



Early treatment = less pain, shorter flare, faster recovery.

2 Rest and elevate the affected joint

Even light pressure — a bedsheet — can be excruciating during an acute flare. Rest the joint, keep it elevated, and apply ice wrapped in a towel for 20–30 minutes several times a day to reduce swelling and pain.



Repeat several times a day.

3 Drink large amounts of water

Aim for 3 liters of water during the flare. Hydration promotes uric acid excretion in urine and dilutes crystal concentration. Avoid alcohol and sugary drinks completely during the attack.



Avoid alcohol and sugary drinks.

4 Continue your urate-lowering medication

If you are already on febuxostat or allopurinol, do NOT stop it during a flare. Stopping and restarting causes uric acid fluctuations that prolong or trigger additional attacks. Keep taking it at the same dose.



Key reminder:

Early action, rest, hydration, and medication adherence are the keys to faster relief and fewer flares.



Act early



Rest & elevate



Hydrate well



Stay on medication

Fig. 5 — Acute gout flare: classic presentation with erythema (redness), warmth, swelling, and exquisite tenderness of the first metatarsophalangeal joint — "podagra" (Greek: "foot trap"). The affected joint is so tender that even the weight of a bedsheet causes severe pain. The ankle, knee, wrist, and elbow are also common sites. Untreated, the inflammation peaks at 12–24 hours and resolves spontaneously in 7–14 days, only to recur with the next trigger.

Medications for Gout — Uric Acid Lowering Therapy (ULT) & Flare Management

Medication	Dose	Use	Key Notes for CKD Patients
Allopurinol	50–300 mg/day	Long-term ULT (first-line)	Start at 50 mg/day in CKD — titrate slowly by 50 mg every 4 weeks to target. Monitor uric acid monthly. DRESS syndrome risk — stop immediately if rash develops (can be life-threatening). HLA-B*5801 testing recommended before starting in Filipino patients. Do not start during an acute flare.
Febuxostat	40–80 mg/day	Long-term ULT (alternative)	Preferred over allopurinol in some CKD patients — does not require renal dose adjustment for mild–moderate CKD. Caution in cardiovascular disease — CARES trial showed increased CV mortality vs allopurinol in patients with established CVD. More expensive than allopurinol.
Colchicine	0.5–1 mg/day	Flare prevention & acute flare treatment	Reduce dose in CKD: 0.5 mg/day if eGFR 30–50; avoid or use 0.3 mg single dose if eGFR <30; generally avoid on dialysis. Use prednisolone instead in severe CKD. Interactions: clarithromycin, cyclosporine, strong CYP3A4 inhibitors — increase colchicine toxicity significantly.
NSAIDs (mefenamic acid, ibuprofen, naproxen)	Standard doses	Acute flare (if eGFR > 30)	Nephrotoxic — avoid in CKD stages 4–5 (eGFR < 30). Reduce renal blood flow → AKI. Even 1–2 doses can precipitate acute-on-chronic kidney injury in vulnerable patients. Short course (3–5 days) may be used if eGFR > 30 with close monitoring.
Prednisolone	20–40 mg/day × 3–5 days	Acute flare (when NSAIDs/colchicine contraindicated)	Effective for acute gout. Use when NSAIDs and colchicine are both contraindicated (e.g., severe CKD on dialysis). Raises blood glucose — monitor closely in diabetics. Taper over 5–7 days to prevent rebound flare. Intra-articular triamcinolone is an option for single-joint flares.

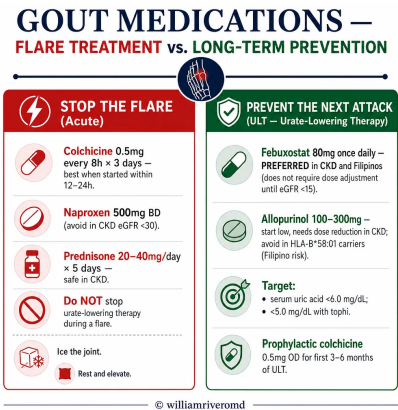


Fig. 6 — Uric acid lowering therapy pyramid: allopurinol is first-line (start low, titrate slowly in CKD); febuxostat for allopurinol-intolerant patients; colchicine for flare prevention and acute treatment with dose reduction in CKD; avoid NSAIDs in eGFR < 30.

How Gout Damages the Kidneys — Three Mechanisms

Mechanism	How it damages the kidney	Clinical consequences
Urate nephropathy	MSU crystal deposits in renal tubules and interstitium → chronic tubular obstruction → interstitial fibrosis → direct CKD progression independent of blood pressure and diabetes	Proteinuria, declining eGFR; most dangerous because it is silent — no joint symptoms required
Uric acid kidney stones	Supersaturation of uric acid in acidic urine (pH < 5.5) → precipitation as uric acid stones → renal obstruction → hydronephrosis → AKI	Flank pain, hematuria, AKI; uric acid stones are radiolucent (invisible on plain X-ray — need CT)
Cardiovascular risk amplification	Hyperuricemia independently associated with hypertension, endothelial dysfunction, and CVD — all of which accelerate CKD progression through reduced renal perfusion	Accelerated CKD progression, increased cardiovascular mortality

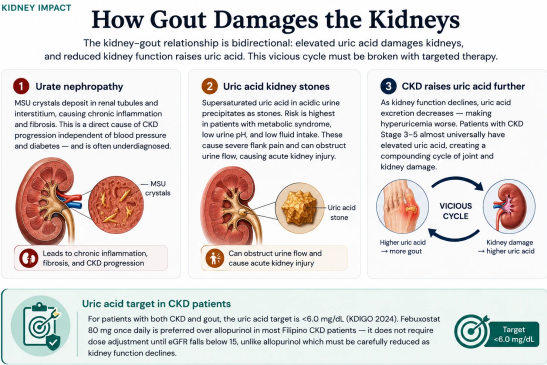


Fig. 7 — Three pathways by which gout damages the kidneys: crystal tubular deposition (urate nephropathy), stone formation and obstruction, and cardiovascular risk amplification. All three are preventable with effective uric acid control.

Long-Term Management Goals

Treatment Targets	Non-Drug Strategies (Adjuncts — Not Substitutes for ULT)
- Serum uric acid < 6.0 mg/dL on treatment (general) - Serum uric acid < 5.0 mg/dL if tophi present or frequent flares - Zero gout flares per year once uric acid is controlled - eGFR stabilization — halt CKD progression	- Hydration: 2–2.5 L water/day (if not fluid-restricted) - Eliminate beer and spirits completely - Avoid organ meats, dillis, sardinas, bagoong - Replace softdrinks with water, plain coffee, unsweetened tea

- Allopurinol/febuxostat: lifelong therapy (stopping causes flares to return)

- Weight loss if overweight — reduces uric acid by 0.02 mg/dL per kg lost
- Coffee 1–2 cups/day — evidence for uric acid lowering
- Low-fat dairy daily — uric acid promotes uric acid excretion

For educational use only. This guide does not replace your physician's advice. Medication dosing must be individualized to your kidney function, medications, and medical history. References: ACR Gout Guidelines 2020 · EULAR Gout Recommendations 2016 · KDIGO CKD Guidelines 2024 · FitzGerald et al., Arthritis Rheumatol 2020 · Richette et al., Ann Rheum Dis 2017.

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